

HW 0428

10.18 Consider the data file *mroz* on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

- Create two new variables. *MOTHERCOLL* is a dummy variable equaling one if *MOTHEREDUC* > 12, zero otherwise. Similarly, *FATHERCOLL* equals one if *FATHEREDUC* > 12 and zero otherwise. What percentage of parents have some college education in this sample?
- Find the correlations between *EDUC*, *MOTHERCOLL*, and *FATHERCOLL*. Are the magnitudes of these correlations important? Can you make a logical argument why *MOTHERCOLL* and *FATHERCOLL* might be better instruments than *MOTHEREDUC* and *FATHEREDUC*?
- Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variable. What is the 95% interval estimate for the coefficient of *EDUC*?
- For the problem in part (c), estimate the first-stage equation. What is the value of the F-test statistic for the hypothesis that *MOTHERCOLL* has no effect on *EDUC*? Is *MOTHERCOLL* a strong instrument?
- Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variables. What is the 95% interval estimate for the coefficient of *EDUC*? Is it narrower or wider than the one in part (c)?
- For the problem in part (c), estimate the first-stage equation. Test the joint significance of *MOTHERCOLL* and *FATHERCOLL*. Do these instruments seem adequately strong?
- For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

d. first stage equation: $EDUC = \gamma_0 + \theta_1 MOTHERCOLL + v$

用OLS估計 first stage equation 取得 fitted value:

$$\Rightarrow \widehat{EDUC} = \hat{\gamma}_0 + \hat{\theta}_1 MOTHERCOLL$$

```
> wage_ols_model <- lm(educ ~ mothercoll, data = mroz1)
> f_test <- summary(wage_ols_model)$fstatistic[1]
> cat("F-test statistic for the hypothesis that MOTHERCOLL has no effect on EDUC: ", f_test, "\n")
F-test statistic for the hypothesis that MOTHERCOLL has no effect on EDUC: 63.21602
```

"F-test statistic = 63.216 > 10

⇒ Hypothesis that IV which MOTHERCOLL is weak is rejected.

e.

```
> mroz_sub <- subset(mroz, wage > 0)
> # 刪除無工資
> mroz(mroz_sub)
[1] 428
>
> iv_model <- ivreg(log(wage) ~ exper+I(exper^2)+educ | exper+I(exper^2)+mothercoll+fathercoll, data = mroz_sub)
> summary(iv_model)

Call:
ivreg(formula = log(wage) ~ exper + I(exper^2) + educ | exper + I(exper^2) + mothercoll + fathercoll, data = mroz_sub)

Residuals:
    min       1Q   median       3Q      max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819   0.3922213  -0.712   0.47714
exper         0.0020261   0.0137950   0.146   0.88433
I(exper^2)    -0.0004848   0.0003976  -1.235   0.03337
educ          0.0878477   0.0007868  111.854   0.00000

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.133, Adjusted R-Squared: 0.147
Wald test: 9.724 on 3 and 424 Df, p-value: 3.224e-06

> confint(iv_model, "educ",
+         level = 0.95)
          2.5 %    97.5 %
educ 0.02751845 0.1481769
```

⇒ 95% CI: [-0.0215, -0.1482]

narrower than the one in part (c)

f. first stage: $EDUC = \gamma_0 + \theta_1 MOTHERCOLL + \theta_2 FATHERCOLL + v$

$$\Rightarrow \widehat{EDUC} = \hat{\gamma}_0 + \hat{\theta}_1 MOTHERCOLL + \hat{\theta}_2 FATHERCOLL$$

```
> # 刪除無工資
> mroz <- subset(mroz, wage > 0)
> # 刪除無工資
> supply(mroz, length)
taxableinc federaltax hsbtlngs hfathereduc hmothereduc siblings 1fp
hours kids16 kids618 age educ wage wage6
hhours hage heduc hwage faminc ntr mothereduc
fathereduc unemployment largcity exper
428 428 428 428 428 428
>
> mroz_lme <- lm(educ ~ mothercoll + fathercoll, data = mroz)
> f_test_f <- summary(mroz_lme)$fstatistic[1]
>
> cat("F-test statistic for joint significance of MOTHERCOLL and FATHERCOLL: ", f_test_f, "\n")
F-test statistic for joint significance of MOTHERCOLL and FATHERCOLL: 57.59666
```

"F-test statistic = 57.596 > 10

⇒ Hypothesis that instruments are weak is rejected.

g. 使用Sargan test 檢驗 overidentifying restrictions 的有效性.

Ho: surplus instruments are valid (和 error 無關)

```
> diag_mat <- summary(iv_model, diagnostics = TRUE)$diagnostics
> q_test <- diag_mat[greg("Sargan"), rownames(diag_mat), "statistic"]
> critical_value <- qchisq(0.95, df=1)
> print(critical_value)
[1] 3.841459
```

```
> cat("Sargan-Hansen statistic for the validity of the surplus instrument: ", q_test, "\n")
Sargan-Hansen statistic for the validity of the surplus instrument: 0.2375851
```

Test statistic 呈 χ^2 分佈.

"5% critical value = 3.841459

⇒ Ho isn't rejected.

⇒ IV (z) is exogeneity from residual ϵ_i .

a.

```
> t1p <- mroz$t1p
> mother_educ <- mroz$mothereduc
> father_educ <- mroz$fathereduc
>
> mothercoll <- ifelse(mother_educ > 12 & t1p == 1, 1, 0)
> fathercoll <- ifelse(father_educ > 12 & t1p == 1, 1, 0)
>
> numbers_mothercoll <- sum(mothercoll)
> numbers_fathercoll <- sum(fathercoll)
>
> percentage_mothercoll <- (numbers_mothercoll / 428) * 100
> percentage_fathercoll <- (numbers_fathercoll / 428) * 100
>
> print(paste("Percentage of mothers with some college education:", percentage_mothercoll, "%"))
[1] "Percentage of mothers with some college education: 22.44932710304 %"
> print(paste("Percentage of fathers with some college education:", percentage_fathercoll, "%"))
[1] "Percentage of fathers with some college education: 11.6822429906542 %"
```

b.

```
> mothercoll <- mothercoll[1 : 428]
> fathercoll <- fathercoll[1 : 428]
> educ <- mroz$educ
> educ <- educ[1 : 428]
> correlation_matrix <- cor(cbind(educ, mothercoll, fathercoll))
> print(correlation_matrix)

      educ mothercoll fathercoll
educ    1.0000000    0.3594705    0.3984962
mothercoll 0.3594705    1.0000000    0.3545709
fathercoll 0.3984962    0.3545709    1.0000000
```

magnitude of correlation: 代表 EDUC 和 parents' education 的關係.

但數據結果卻是中等程度的相關性.

MOTHERCOLL 和 FATHERCOLL 分別為 0.3595 和 0.3985

是較好的 IV, "把父母教育簡化成是否有大學學歷的=元變數,

減少連續年數的極端值影響, 且符合外生性.

c. $\lambda_n(WAGE) = \beta_0 + \beta_1 EXPER + \beta_2 EXPER^2 + \beta_3 EDUC + \epsilon$

$$\Rightarrow \lambda_n(WAGE) = \beta_0 + \beta_1 EXPER + \beta_2 EXPER^2 + \beta_3 MOTHERCOLL + \epsilon^*$$

first stage: $EDUC = \gamma_0 + \theta_1 MOTHERCOLL + v$

second stage: $\lambda_n(WAGE) = \beta_0 + \beta_1 EXPER + \beta_2 EXPER^2 + \beta_3 MOTHERCOLL + \epsilon^*$

```
> mroz1 <- mroz[mroz$t1p==1,]
> mroz1$MOTHERCOLL <- ifelse(mroz1$mothereduc > 12, 1, 0)
>
> wage_iv_model <- ivreg(log(wage) ~ educ + exper + I(exper^2) | mothercoll + exper + I(exper^2), data = mroz1)
> educ_95percent_interval <- confint(wage_iv_model, level = 0.95)("educ",)
> confint(wage_iv_model, "educ", level = 0.95)
          2.5 %    97.5 %
educ -0.001219763 0.1532557
```

⇒ 95% CI: [-0.00122, 0.15323]

10.20 The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where r_j and r_f are the returns to security j and the risk-free rate, respectively. r_m is the return on the market portfolio, and β_j is the j th security's "beta" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

- Use the observations on Microsoft in the data file `capm5` to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?
- It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \dots, 180$. Does this value potentially satisfy the conditions IV1-IV3? Create $RANK$ and obtain the first-stage regression results. Is the coefficient of $RANK$ very significant? What is the R^2 of the first-stage regression? Can $RANK$ be regarded as a strong IV?
- Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?
- Use $RANK$ as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?
- Create a new variable $POS = 1$ if the market return $(r_m - r_f)$ is positive, and zero otherwise. Obtain the first-stage regression results using both $RANK$ and POS as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?
- Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?
- Obtain the IV/2SLS estimates of the CAPM model using $RANK$ and POS as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?
- Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

a.

```
> capm5msft_excess = capm5msft - capm5riskfree
> capm5mkt_excess = capm5mkt - capm5riskfree
> model_ols = lm(msft_excess ~ mkt_excess, data = capm5)
> summary(model_ols)

Call:
lm(formula = msft_excess ~ mkt_excess, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27424 -0.04744 -0.00820  0.03869  0.35801

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003250   0.006036   0.538   0.591
mkt_excess  1.201840   0.122152   9.839 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
Multiple R-squared:  0.3523, Adjusted R-squared:  0.3486
F-statistic: 96.8 on 1 and 178 DF, p-value: < 2.2e-16
```

p -value 很小 $\Rightarrow$ relationship 是顯著的.

$\beta > 1 \Rightarrow$ Microsoft 比市場有風險, 比CAPM模型認為更有風險.

b.

```
> capm5RANK <- rank(capm5mkt_excess, ties.method = "first")
> first_stage <- lm(mkt_excess ~ RANK, data = capm5)
> summary(first_stage)

Call:
lm(formula = mkt_excess ~ RANK, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.110497 -0.006308  0.001497  0.009433  0.029513

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.903e-02  2.195e-03  -36.0 <2e-16 ***
RANK        9.067e-04  2.104e-05  43.1 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126, Adjusted R-squared:  0.9121
F-statistic: 1858 on 1 and 178 DF, p-value: < 2.2e-16
```

$RANK$ 對 Microsoft 的報酬率並沒有顯著影響, $\Rightarrow$ 滿足 IV1

regression error in time t 有可能和 ranked value 無相關性 $\Rightarrow$ 有可能滿足 IV2

$r_m - r_f$ 較大時, $RANK$ 變大 $\Rightarrow$ 滿足 IV3

R^2 和 F 值都很大. $\Rightarrow RANK$ 若為 valid, 則是個很強的 IV.

C.

```
> capm5v_hat <- residuals(first_stage)
> model_augmented <- lm(msft_excess ~ mkt_excess + v_hat, data = capm5)
> summary(model_augmented)

Call:
lm(formula = msft_excess ~ mkt_excess + v_hat, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27140 -0.04213 -0.00911  0.03423  0.34887

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.005984   0.504   0.6146
mkt_excess  1.278318   0.126749  10.085 <2e-16 ***
v_hat       -0.874599   0.428626  -2.040   0.0428 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672, Adjusted R-squared:  0.36
F-statistic: 51.34 on 2 and 177 DF, p-value: < 2.2e-16
```

是檢驗 endogeneity of the Hausman test.

$\hat{v}$ 的 t -statistic $= -2.040$, p -value $= 0.0428$, 在 1% 的水準不顯著.

$\Rightarrow$ 無法拒絕 market return 是 exogenous 的無偏假設.

d.

```
> model_iv <- ivreg(msft_excess ~ mkt_excess | RANK, data = capm5)
> summary(model_iv)

Call:
ivreg(formula = msft_excess ~ mkt_excess | RANK, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.271625 -0.049675 -0.009693  0.037683  0.355579

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.006044   0.499   0.618
mkt_excess  1.278318   0.128011   9.986 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-squared:  0.3508, Adjusted R-squared:  0.3472
Wald test: 99.72 on 1 and 178 DF, p-value: < 2.2e-16
```

IV/2SLS 的 $\beta = 1.278318$ 比原 OLS 的 1.201840 稍大

若有 measurement error 的問題, OLS estimator 的 bias 會向下偏斜.

$\Rightarrow$ 符合預期.

e.

```
> capm5POS <- ifelse(capm5mkt_excess > 0, 1, 0)
> first_stage2 <- lm(mkt_excess ~ RANK + POS, data = capm5)
> summary(first_stage2)

Call:
lm(formula = mkt_excess ~ RANK + POS, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.109182 -0.006732  0.002858  0.008936  0.026652

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0804216  0.0022622  -35.55 <2e-16 ***
RANK        9.0009819  0.0000400  24.55 <2e-16 ***
POS        -0.0092762  0.0042156  -2.20   0.0291 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149, Adjusted R-squared:  0.9139
F-statistic: 951.3 on 2 and 177 DF, p-value: < 2.2e-16
```

F -statistic $= 951.3 \Rightarrow$ 由 F -test 可知 IV 不弱

$R^2 = 0.9149 \Rightarrow RANK$ 是強烈顯著, 但 POS 在 1% 的水準不顯著.

但在 5% 水準顯著, 則可說 IV 不弱.

f.

```

> first_stage3 <- lm(mkt_excess ~ RANK + POS, data = capm5)
> capm5$v_hat2 <- residuals(first_stage3)
> model_hausman <- lm(msft_excess ~ mkt_excess + v_hat2, data = capm5)
> summary(model_hausman)

Call:
lm(formula = msft_excess ~ mkt_excess + v_hat2, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27132 -0.04261 -0.00812  0.03343  0.34867

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004   0.005972   0.503   0.6157
mkt_excess   1.283118   0.126344  10.156 <2e-16 ***
v_hat2      -0.954918   0.433062  -2.205   0.0287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696,    Adjusted R-squared:  0.3625
F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16

```

first stage residual 的 t -value = -2.205, p -value = 0.0287, 在 1% 水平不顯著。

⇒ 無法拒絕 market return 是 exogenous 的虛無假設。

g.

```

> model_iv2 <- ivreg(msft_excess ~ mkt_excess | RANK + POS, data = capm5)
> summary(model_iv2)

Call:
ivreg(formula = msft_excess ~ mkt_excess | RANK + POS, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27168 -0.04960 -0.00983  0.03762  0.35543

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004   0.006044   0.497   0.62
mkt_excess   1.283118   0.127866  10.035 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-squared:  0.3507,    Adjusted R-squared:  0.347
Wald test: 100.7 on 1 and 178 DF,  p-value: < 2.2e-16

```

IV coefficient estimate = 1.283118 比 OLS estimate = 0.201840 稍大

⇒ 符合預期。

h.

```

> model_iv3 = ivreg(msft_excess ~ mkt_excess | RANK, data = capm5)
> uhat = resid(model_iv3)
> sargan_test = lm(uhat ~ RANK + POS, data = capm5)
> summary(sargan_test)

Call:
lm(formula = uhat ~ RANK + POS, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.26915 -0.04698 -0.00798  0.03771  0.35670

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0026225   0.0126308  -0.208   0.836
RANK         0.0001417   0.0002233   0.635   0.526
POS         -0.0174944   0.0235375  -0.743   0.458

Residual standard error: 0.08102 on 177 degrees of freedom
Multiple R-squared:  0.003111, Adjusted R-squared: -0.008153
F-statistic: 0.2762 on 2 and 177 DF,  p-value: 0.759

>
> n = nrow(capm5)
> R2 = summary(sargan_test)$r.squared
> S = n * R2
> S
[1] 0.5600477 → test statistic.
> qchisq(0.95, df = 1) → rejection region:  $\chi^2 > 3.841459$ 
[1] 3.841459

```

⇒ 無法拒絕 $\text{cov}(RANK, e) = 0$, $\text{cov}(POS, e) = 0$ 的虛無假設

⇒ RANK 和 POS 是有效的 IV。